

Reimagining Recovery: A Patient-Centred In-Hospital Intervention to Motivate Early Mobilisation Post-Surgery

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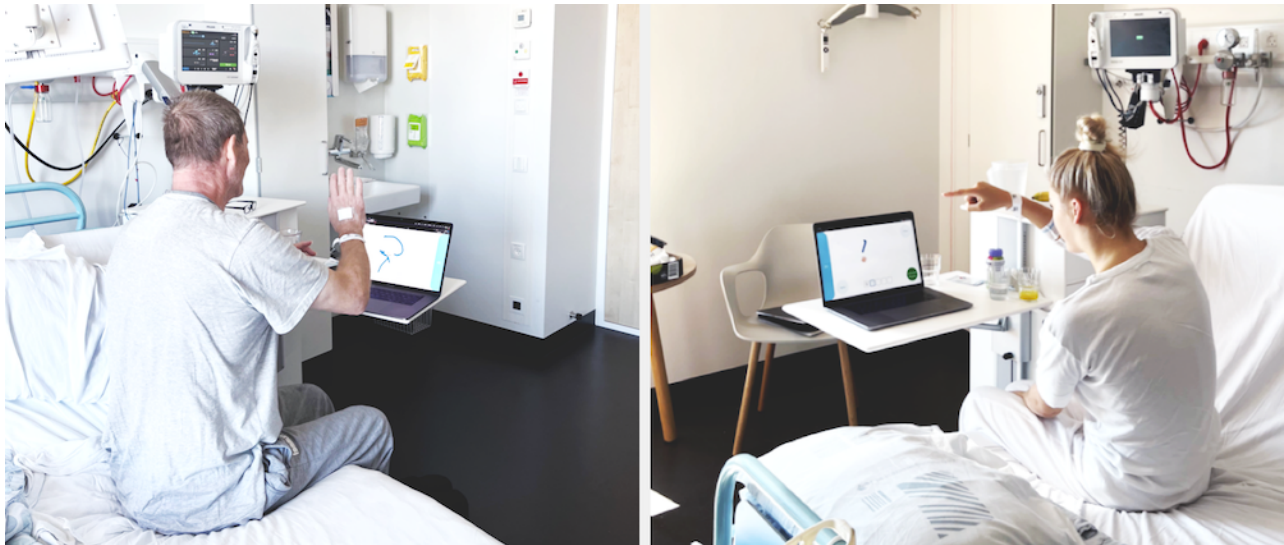


Figure 1: Patients at the Gastroenterology department at a major Danish university hospital playing with Kinetip

Abstract

A lack of physical activity post-surgery is widely acknowledged as both a significant barrier to successful recovery and contributor to post-operative complications. In Denmark, the recent introduction of single-patient hospital rooms — intended to improve care — has

unintentionally reduced patients' early mobilisation, potentially due to a lack of tacit social and other motivations. While prior HCI research has explored design for motivation, rehabilitation, and physical activity, limited attention has been granted to understanding the factors and practices that influence these behaviours in this specific clinical context. This paper presents a mixed-methods, user-centered design research process leading to the development of a novel gamified, digital solution to motivate patients to engage in physical activity following gastroenterology surgery. Through this work, we offer initial evidence for the feasibility of this system based on a preliminary evaluation in-clinic ($n = 16$), and offer further implications for the design of patient-centred digital rehabilitation therapy supports.

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CCS Concepts

• Human-centered computing;

Keywords

Human-Computer Interaction, Hospital, Clinic, Gastroenterology, Surgery, Rehabilitation, Physical Therapy, Gamification, Patient-Centered Care, Human-Centered Design, Hospital Room, Motivation, Gesture-Based Interaction

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1 Introduction

The days, weeks, and months following any surgical procedure are essential to patients' short-term recovery and long-term outcomes [43]. Early Mobilisation (EM), in particular, has a significant impact on the health of hospitalised patients, yielding benefits including a reduction in the duration of hospital stay, acceleration of recovery and rehabilitation, and improvement of physical function [7, 11, 33, 80]. Given its importance, EM has been widely studied for many years [46], producing a rich literature concerning its likely causes and resolutions, from personalised rehabilitation therapies [67, 68] to revised hospital cultures [25, 35]. Despite this extensive prior research, sedentary behaviour yet remains common during hospital stays, regardless of the primary cause for admission [7, 11]. Hospitalised patients generally spend most of their time in bed [11], and rely on physiotherapists to communicate and encourage their adherence to often complex behavioural change models [56].

Denmark in particular has in recent years launched initiatives to improve post-surgical care [63, 71, 72], including modernising hospital facilities [37]. At a major Danish university hospital, such modernisation efforts have resulted in single-occupancy rooms in place of the double rooms previously the norm; as means to improve privacy, comfort, and infection control [62]. Within the Gastroenterology department, this change has yet led to an unexpected, adverse effect — healthcare professionals observing a further decline in hospitalised patients' physical activity (PA). This has serious implications for patients' wellbeing and recovery [7, 11, 33], and highlights the challenge of ensuring optimal conditions for post-surgery mobilisation in the Danish healthcare system.

Many methods have been developed to promote early mobilisation among patients after surgery [26]. Measures to increase PA among patients which require the support of hospital staff are yet time-intensive and costly [12, 14, 47]. For this reason, many have recently turned towards the potential of digital technologies to support rehabilitation in a more cost-effective and yet caring fashion [2, 6, 9, 48, 66, 82]. The successful adoption of such systems in the clinical context, has yet proved largely limited, a result it has been argued of “the full scope and intricacy of the problem space” [41]. This, despite calls to indeed leverage the potential of technology to permit increased physicality and interactivity [48].

This is a particularly interesting challenge for Gastroenterology patients who commonly present symptoms such as abdominal pain, nausea and fatigue, experience short-duration hospital stays of up to three days, and often go through surgery during this period. In 2022 in Denmark, 74% of the patients admitted for gastroenterology-related diagnoses were 45 years of age or older, making the average age of a patient in the Gastroenterology department 60 years old [1]. To date, surprisingly few efforts have yet been made to engage patients and clinicians in the user-centred design of digital rehabilitation technologies [3, 45], leaving a gap in knowledge concerning the design and adoption of effective in-hospital PA interventions. This includes to facilitate relatively independent use [21], to motivate physical activity after surgery [3, 5, 9, 38, 81], and to balance medical and patient needs [45]. Even fewer are the number of systems brought beyond prototypes to the point of implementation and evaluation in the hospital context [3, 74].

We recount in this paper a programme of mixed-methods design research conducted in parallel with a full technology development and evaluation lifecycle; comprising research phases including research activities conducted with hospitalised patients and staff (n=11), ideation and design workshops (n=27), the development of a novel digital solution, and a preliminary evaluation with patients in-clinic (n=16). We undertook these efforts to address the gaps in the current HCI for rehabilitation literature, as to promote inpatient mobilisation through a technological intervention to motivate PA among patients in single-bed hospital rooms. Through these efforts, we contribute to the CHI community and literature: a) an in-depth analysis of the factors contributing to low physical activity levels among hospitalised patients in a gastroenterology ward, and b) implications for the design of digital rehabilitation technologies, realised in the form of a bespoke digital tool for in-patient mobilisation.

2 Related Work

2.1 Designing for Physical Rehabilitation

Significant evidence is offered by the medical literature for the positive impact of early mobilisation during the perioperative period on patients' recovery and subsequent duration of hospital stay [8]. Such findings have motivated the development of guidelines including the Enhanced Recovery after Surgery System (ERAS) [49], defining an evidence-based pathway for recovery including emphasising to patients the importance of early mobilisation [26]. There remains however, a widely-acknowledged series of barriers to patient engagement in physical activity, spanning medical and social factors as well as beliefs about consequences [28] — which others yet argue can be overcome by positive social influences, goal-setting, and behaviour regulation activities [76]. These are each domains of human endeavour which it is believed digital technologies present great promise to support, as evidenced by calls not only from researchers [4] but institutions such as the Danish Ministry of Health (Sundhedsministeriet) which has recently advocated for the increased digitisation of national patient services [72].

Design for patient mobilisation and physical rehabilitation has indeed developed over the past decade into a significant HCI research thread. Much of this recent prior work has focused on leveraging advanced technologies for analysing human motion

[2, 12, 48, 55, 82], including wearables, machine learning techniques, robots, virtual reality (VR) treadmills, and leap-motion controllers. Many related innovations have been inspired by efforts to shift physical rehabilitation activities from the clinic to the at-home context [6, 9, 14, 18, 38, 45, 50, 52, 65, 77], or to leverage technology to support longitudinal rehabilitation [3, 41, 42, 66, 74]. While much innovative HCI research has therefore been conducted with respect to physical rehabilitation, it is worth noting that many such efforts either require the adoption of advanced, emergent technologies [9] or remain at an early design stage [77], still yet to yield evidence for the real-world engagement of users.

2.2 Designing for Motivation

Motivation is widely acknowledged as a crucial factor in successful rehabilitation therapy [30], and yet existing support systems have equally been criticised for the extant challenge of measuring their success, the burden on clinicians, and for simply being “boring” [75]. This, despite much HCI research devoted to understanding the motivations of diverse users of technology across numerous domains [22, 24, 54, 60] including games and physical activity applications, and embracing a wide range of theories [19, 44, 58, 60, 64] including most prominently Self-Determination Theory (SDT), Flow Theory and diverse behaviour change theories. Prior HCI research on physical rehabilitation in particular has highlighted the importance of designing for patient motivation [10, 16, 45, 78], employing strategies from framing technologies as ‘motivational aids’ to offering ‘rewarding feedback’, adaptive achievements and personalised PA recommendations.

Such works have also spoken to the value of playful approaches to enabling successful digitally-mediated rehabilitation therapies, including by attending to humour, novelty, and even cosiness [3, 9, 18, 51]. Prior work has focused on leveraging game-like features to “tap into users’ inherent desire for achievement and progress, making rehabilitation exercises more interactive and enjoyable” [5, 50, 52], explored the potential of emerging VR and wearable technologies for the gamification of the rehabilitation process for at-home user engagement [52], highlighting in particular the importance of striking an appropriate balance between in-game challenge and patient motor skill [5]; and has yet emphasised the importance of attending to motivational factors in physical therapy beyond balancing degrees of skill and challenge [5], including situation-specific factors, individual traits and the felt dimension of movement-based interaction design [6, 34]. Prior HCI research has then showcased the potential of mixed-media technologies to increase patient engagement [45, 81] and at the same time the need to move beyond the limitations of typical rehabilitation therapies by leveraging the potential of technology to permit increased physicality and interactivity [48].

2.3 Designing for the Hospital Room Context

Meeting such calls to take into account “the full scope and intricacy of the problem space” of physical rehabilitation technology development [41], then requires also, careful consideration of the hospital patient room, a surprisingly complex and under-explored context for design. That prior HCI research which has turned its attention

to the hospital-room as design context [10, 20, 21, 32, 79], highlights the significance of diverse challenges including managing time amidst constant change [32, 59], hospital workflows, privacy and security concerns [20], and the importance of maintaining a sense of the hospital as a trusted place [21] as well as an overview of the complete patient journey [36].

Prior HCI research in the hospital context has then emphasised the importance of involving therapists in the early stages of design to permit greater user engagement in the longer-term [45], attending to diverse considerations including cost as a barrier to the adoption of rehabilitation solutions [12], mitigating the need for maintenance [47], and supporting a yet “relaxing, entertaining and immersive rehabilitation atmosphere” [3]. These challenges overcome, prior research highlights the positive potential impact of employing innovative digital technologies [15, 65] within patient rooms on patient wellbeing; by augmenting patients’ sense of independence, autonomy and control, as well as enhancing feelings of at-homeness [21], while yet producing new calls for HCI researchers to move beyond leveraging digital technologies for basic room control to realising their potential to support therapy and rehabilitation [20]. There then remains a need to develop new knowledge through design of the complexities of integrating digital rehabilitations systems into the day-to-day context of the single-bed hospital room in which reside patients with limited mobility [23]. To address this need, we take inspiration from a prior call by HCI researchers working within the unique context of the hospital patient room to “design to augment” yet “not interfere” [59].

3 Method

We then in this work adopted an interdisciplinary design research approach, following the Double Diamond model [17], to explore the antecedents of effective human-computer interaction for rehabilitation. A total of 54 participants were involved throughout this iterative process, including 4 nurses, 2 physiotherapists, and 48 patients. Participants were recruited through collaboration with a Professor of Clinical Nursing at the Gastroenterology Department of a major Danish university hospital, who facilitated access to this clinical context. While formal ethical approval was not required for this work according to the established standards of this context, we adhered to the highest ethical standards. All data was managed in compliance with relevant legislation, including the General Data Protection Regulation (GDPR). Participants received an information sheet in line with university policies and provided signed consent before participation.

This work was carried out from February to July 2022 and consisted of four key phases, followed by a preliminary evaluation. The first phase involved gaining insight into the experiences of patients and staff across three days of participant observations conducted during the evening shift on the Gastroenterology ward, where the research team shadowed a nurse or assistant. This allowed us to form a preliminary understanding of the problem space while minimising preconceived biases. Following these observations, eight semi-structured interviews and focus groups were conducted with stakeholders, including 5 patients, 4 nurses, and 2 physiotherapists. These interviews were recorded, transcribed and translated from Danish for further analysis in the second phase; by reflexive

thematic analysis, following Braun and Clarke [13], of the interview and focus group data. These same data were also subject to a Root Cause Analysis with an explicit focus on the pillars of Self-Determination Theory [27, 61]. The themes and causal factors identified by these methods were then compared, combined and refined to form *five actionable requirements* to pursue in the next phase, as further described in Section 5.

In the third phase, we explored diverse possible solutions aligned with these actionable requirements through idea generation and conceptualisation methods, resulting in two concepts selected for further analysis. To analyse these concepts, we adopted the principles of rapid prototyping to attain efficient input and feedback from target users. The concepts were validated over two days with 7 patients through a Wizard of Oz approach [31], with a focus on participants' engagement and the appeal of the concepts. Participants were introduced to each concept, and their comments and observations were recorded. Analysis, alongside a SWOT assessment, led to the selection of Concept B for further exploration in the fourth phase. This phase consisted of identifying and further developing a single solution, by iterating upon and refining this chosen concept. In between iterations, we continually revised a prototype system, by incorporating users' feedback into the design loop. Using Think Aloud [39] and System Usability Scores [40] as methods, a usability test was conducted over three days, involving 7 and 13 participants respectively. This phase yielded a web-based puzzle game with which users would interact through movements detected by computer vision; implemented using Hand-tracking [57], and Optical Character Recognition machine learning models [73].

Finally, we carried out a preliminary evaluation with 16 participants of this functioning prototype; which held as its objective, developing an initial sense of the potential feasibility of this system for motivating physical activity in-hospital. The evaluation was conducted over two consecutive days, each session lasting around 30 minutes per participant. Following informed consent, participants first completed a questionnaire containing demographic questions. They were then introduced to the solution and played the game while the authors were present. Finally, participants responded to a questionnaire containing a short-form version of the Intrinsic Motivation Inventory (IMI) [27]. The IMI allowed us to assess via a score from 1 to 7, participants' *enjoyment* (as a measure of intrinsic motivation), *perceived choice* and *perceived competence* (both indicators of self-regulated motivation), and *pressure* (a negative predictor of self-regulated or intrinsic motivation) [27], serving overall as an indicator of motivation to engage in physical activity when interacting with the platform.

4 Findings

4.1 Single-Occupancy Room Patients Often Feel Uninspired

Hospitalised patients in the Gastroenterology ward were mostly over 50 years old, with stays averaging up to three days. Conversations revealed patients' desire to regain physical abilities and return home. However, the inability to fulfil these wishes, coupled with the experience of being hospitalised, had a clear psychological impact which, in many cases, influenced patients' self-perceptions;

“when you're here, you're sick, right? And when you are at home you are not sick. So you also lack some freedom that is taken from you” (P1). Our observations also confirmed that patients indeed spent most of their time in a sedentary state. Despite healthcare professionals' efforts to encourage movement, nurses often noted resistance from patients; *“It is not enough just one push, I have to go there like many times ... until they said yes”* (N1). When patients did move however, it was largely due to practitioners' persistence. This external motivation contrasts with patients' lack of intrinsic motivation, explained by fear of injury, a lack of confidence, need for assistance, and unclear guidance as to safe movement. These insights highlight the need to better address patients' psychological needs, fostering enjoyment and independence in physical activity through encouragement.

4.2 There Exists Widespread Uncertainty as to Which Forms of Physical Activity Matter and Are Safe

When asked about physical activities during hospitalisation, some patients mentioned walking, one noting, for example; *“the other day I was out walking around the entire hospital, it was 2.5 km”* (P1). Such comments reflect a tendency among patients to view physical activity through a conventional lens, and yet nurses and physiotherapists often noted that, for many, PA tied to daily tasks, like hygiene routines, was more appropriate for rehabilitation. Misconceptions about meaningful physical activity could be problematic for patients, as vigorous goals could come to feel unattainable, lowering perceived competence and reducing motivation. This was evident when patients cited barriers to physical activity, one noting; *“I can't move enough ... Otherwise, I would jump out of bed.”* (P4). Additionally, some patients dismissed the importance of early mobilisation; *“I'm here for such a short period, so I don't feel the need for that ... I will just take it easy”* (P3). Despite professionals' efforts to stress the importance of post-operative activity, this mindset was common. Understanding these perspectives and motivations is crucial, especially when addressing older adults whom tend to value the restorative function of sedentary activities, and for some among whom, health is no longer an important indicator of QoL.

4.3 The Hospital Environment Matters

When asked about enjoyable aspects of their hospital stay, all patients praised the staff's kindness and attentiveness. However, patients also reported that the healthcare practitioners were busy and did not want to disturb them unnecessarily; *“You cannot keep pressing the red button, right? You know that they are busy”* (P1). These feelings made patients hesitant to rely on nurses, unsure of receiving the necessary support. We furthermore observed that the hospital's organisational structure created bottlenecks for staff, as nurses' increased taskload led to the down-prioritization of other responsibilities. As one of the nurses noted; *“We have busy days, so there's a limit to how many times we can motivate the patients and ask them for a walk”* (N3). This barrier affects not only patients needing assistance with movement but also those who, though willing to move, are uncertain of what to do or where to go.

At this major Danish university hospital, single-bed rooms are equipped with all necessary amenities — including a bed, tablet,

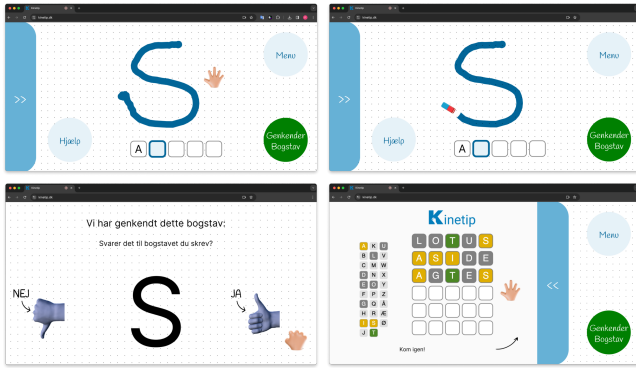


Figure 2: The Kinetip Interface

seating area, and private bathroom — and although this increases patient comfort, it was also found to discourage physical activity, given little need to leave the room. This, coupled with the lack of resources and space to support physical activity within and outside the room, often led to patients’ self-isolation and sedentary behaviours, fostering a diminished sense of relatedness; *“it can also be lonely ... in some hospitals they have made sliding doors, so if you want, you can open up to the next patient, and that’s smart”* (P1).

4.4 Assessment of Physical Activity Is Often Ad-Hoc

During three days of observations, we noted healthcare professionals’ efforts to encourage patient movement, yet little attention was given to assisting with mobility or tracking progress. Interviews revealed that motivating patients was a task shared between nurses and physiotherapists, however responsibility for mobilisation and progress monitoring remained unclear. Such unclear task division could then lead to overlooked duties, and task overload to communication challenges. Physiotherapists articulated their mobilisation responsibilities as including assessing patients’ physical states and recommending post-surgery movements, and yet most rehabilitation sessions took place after discharge. As for the task of tracking progress, both nurses and physiotherapists reported inconsistent methods of measuring PA; *“Yeah, there is no measuring ... It’s random actually, it’s not that standardised”* (N1). This lack of proper tools and standardised practices for measuring physical activity hinders data documentation and sharing, leading to task repetition for nurses and limiting personalised mobilisation support, making progress tracking challenging.

5 Design & Evaluation

Given the lack of perceived value associated with physical activity and the absence of a felt sense of autonomy, competence, and relatedness among patients, we focused in this work on strategies to enhance their intrinsic motivation [64]. While fully fostering intrinsic motivation during hospitalisation may not be realistic, we felt certain actions could be taken to begin supporting these needs. We chose to incorporate extrinsic motivators, such as tangible rewards, social recognition, and measurable goals, as these are often effective in contexts requiring immediate engagement. Based on

this approach, we devised the following design requirements to guide our development of a solution which must (i) appeal to the target group, primarily older adults with gastrointestinal complications; (ii) be accessible for patients hospitalised in single rooms; (iii) facilitate appropriate physical activity with minimal aid from healthcare practitioners; (iv) visualise progress to permit users to track their self-development; and (v) create achievable goals based on individual performance.

5.1 The Solution: Kinetip

Through this process, we arrived at a final solution in the form of a digital game platform we call *Kinetip*, a web-based solution accessible online at kinetip.dk (See Figure 1 and Figure 2). This system is designed to support a variety of games through gesture-based control; encouraging users to move their hands and arms while playing and interacting with the hospital terminal device. Time constraints, however, meant that we had to limit the implementation of *Kinetip* in this first instance to a single game; and opted for a word puzzle game, permitting the combination of cognitive and physically-engaging activities. This game draws inspiration from Wordle, an existing puzzle game with remarkable popularity in recent years [70], particularly among older populations [69]. In this game, users strive to guess a random word within a fixed number of attempts. While adhering to the rules and dynamics inherent to Wordle, the aim was to leverage the existing engagement afforded by the game to enable a context requiring users to move their bodies. Through *Kinetip*, instead of entering the letters through the keyboard, users articulate their guesses by employing their arms and hands to draw each letter of the word. By gamifying the experience of physical activity and leveraging established drivers of behaviour change through gesture-based interaction, we sought to render the mundane task of being physically active while hospitalised more enjoyable.

5.2 Evaluation of Kinetip

A preliminary evaluation of *Kinetip* provided new insight into the potential of such technology to enhance in-patients’ motivation for physical activity and to inform its iterative design. The results of the IMS revealed fairly high ratings for *Kinetip* along the dimensions of interest/enjoyment and perceived choice, with scores, out of seven, of 5.43 and 5.87 respectively (See Table 1); suggesting patients were motivated by the fun or challenge of the activity. The perceived competence score was however moderate, at 4.33 out of seven, hindering us from confirming that integrated regulation was reached. We furthermore identified a positive correlation ($p=0.044$) between perceptions of choice and enjoyment, which are indicators of intrinsic motivation. Results also showed a strong inverse correlation ($p=0.002$) between the pressure and perceived competence experienced while playing the game, indicating that the more pressure the participants felt, the less competence they experienced. Overall, although these results showed slight indications for intrinsic motivation, a more extensive study would be encouraged to determine the lasting impact of this system.

Table 1: Left: Mean and Standard Deviation (SD) of the IMI sub-scores (ranging from 1 to 7) of Interest/Enjoyment (Int.), Perceived Competence (Comp.), Perceived Choice (Cho.), and Pressure/Tension (Pre.). Right: The correlation matrix for each of the sub-scales.

Sub-scale	Mean	SD	Int.	Comp.	Cho.	Pre.
Interest	5.43	0.99	1			
Competence	4.33	1.23	0.51	1		
Choice	5.87	1.26	-0.26	0.19	1	
Pressure	2.92	0.87	-0.23	-0.71	-0.34	1

6 Discussion

Through this work, we highlight how patients rely on external factors for their engagement in physical activity — factors which ultimately necessitate personalised assistance to build confidence. This is often made challenging by the constraints of the secondary care context, in response to which we offer a digital platform, *Kinetip*, to enhance patients' independent yet context-appropriate, and, importantly, enjoyable engagement in early mobilisation. We conclude this paper by offering a series of implications for the design of digital rehabilitation technologies.

6.1 Leverage the advantages of digital technologies to connect hospital, home, patients and professionals

Developing digital platforms for rehabilitation offers unique advantages in meeting accessibility and scalability requirements [20, 21]. Ensuring patients can easily access a platform — whether through hospital-provided tablets or personal devices — extends its reach beyond the hospital setting, allowing rehabilitation to continue at home. In this way, *Kinetip* fosters long-term engagement and adherence to rehabilitation routines while ensuring continuity of care. This is achieved by employing engaging digital affordances as a means to potentially foster the critical therapist-patient relationship [14, 18], as well as to involve other caregivers [14]. Digital technologies such as *Kinetip* can enable personalised rehabilitation by collecting and utilising patient data to set individualised goals based on performance, enhance motivation, and visualise progress to help users track their development effectively, while at the same time providing proof of care and patient progress [14].

6.2 Make movement implicit through gesture

Gesture-based interaction in rehabilitation platforms offers a flexible means to keep patients active without rigid exercise routines. In *Kinetip*, many gestures align with natural communication patterns, making them intuitive and easy to learn. This approach promotes safe PA, even for patients in bed, with minimal assistance. Gesture-based interaction also leverages gamification to motivate users. This, in line with prior design recommendations, including to facilitate the exploration and practice of specific movements, work with movement quality, explore diverse modes of social play, develop concentration and a focus on movement, and foster feelings of success [53, 81]. By requiring hand and arm movements to perform

actions, users' inputs become a unique reflection of their physical effort and coordination. This fosters a sense of ownership and personal development, key drivers of sustained engagement and motivation.

6.3 Employ gamification strategies aligned with familiar practices

Integrating gamification strategies into rehabilitation offers a practical way to engage diverse target groups, including older adults with specific medical conditions [5]. An effective approach is to build upon the target audience's existing interests rather than creating motivation from scratch. For example, leveraging familiar game genres like word puzzles — particularly popular among older adults [69] — helps *Kinetip* establish an immediate connection that fosters curiosity. Cognitive games in particular can serve a dual role by training not only the body but also the mind — in line with prior research findings as to the increased positive cognitive health impact of combined physical and cognitive exercises compared to cognitive exercises alone [29]. Future iterations could further enhance engagement by incorporating a wider variety of game options, genres, and social features to better motivate patients. For instance, *Kinetip* could introduce a collaborative component, wherein patients on the ward work together to guess words within a shared phrase, fostering social interaction and a sense of community in rehabilitation.

7 Conclusion

Through this work, we have explored the design space for physical rehabilitation therapy supports tailored to in-hospital and post-surgery contexts. We offer initial evidence for their feasibility, new knowledge of the needs and aspirations of patients and clinicians, and further implications for the design of digital rehabilitation technologies, which we aspire to see increasingly employed in practice.

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